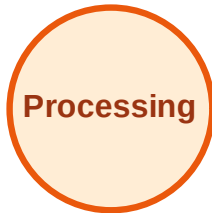


BFS Traversal

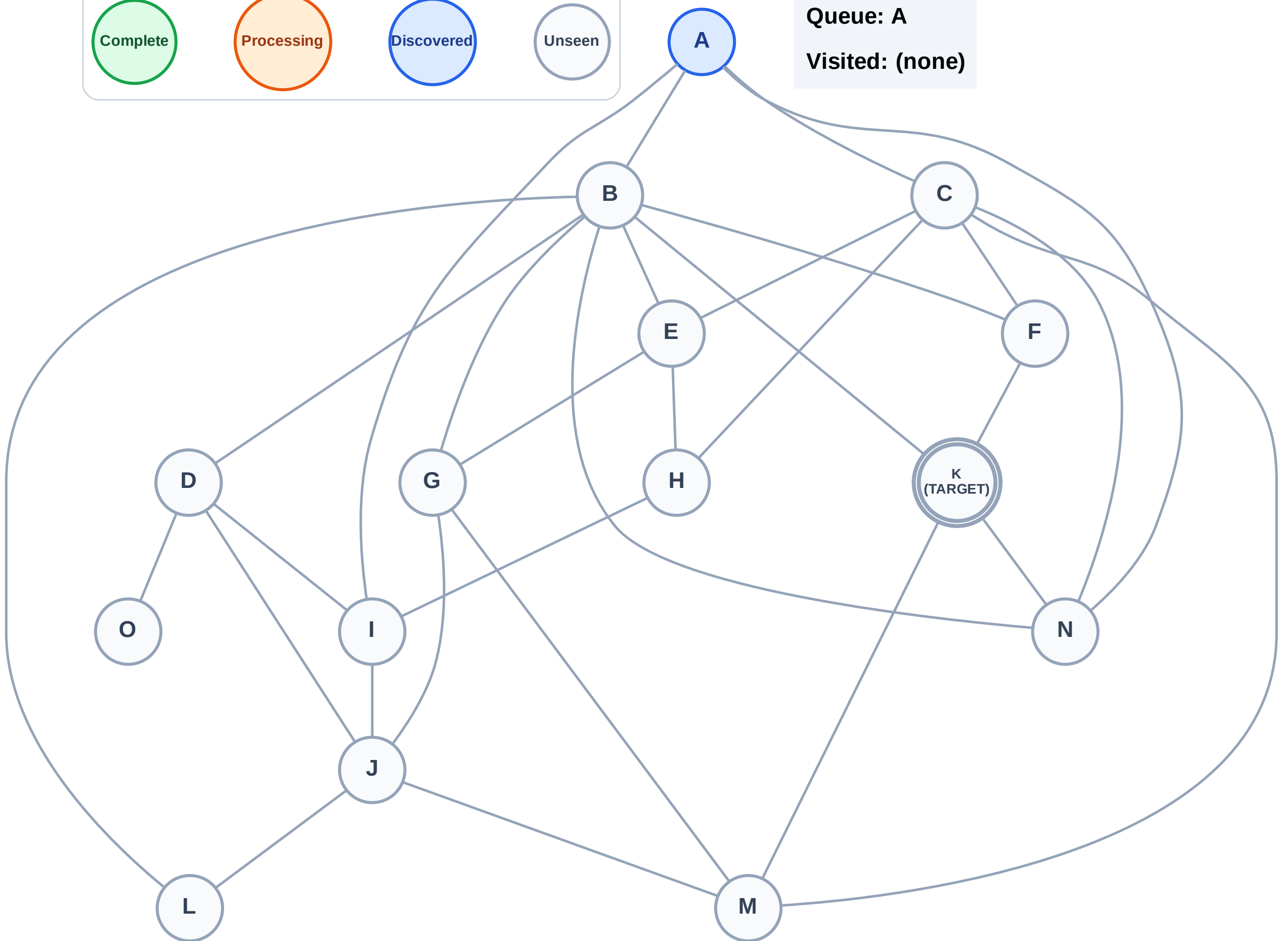
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

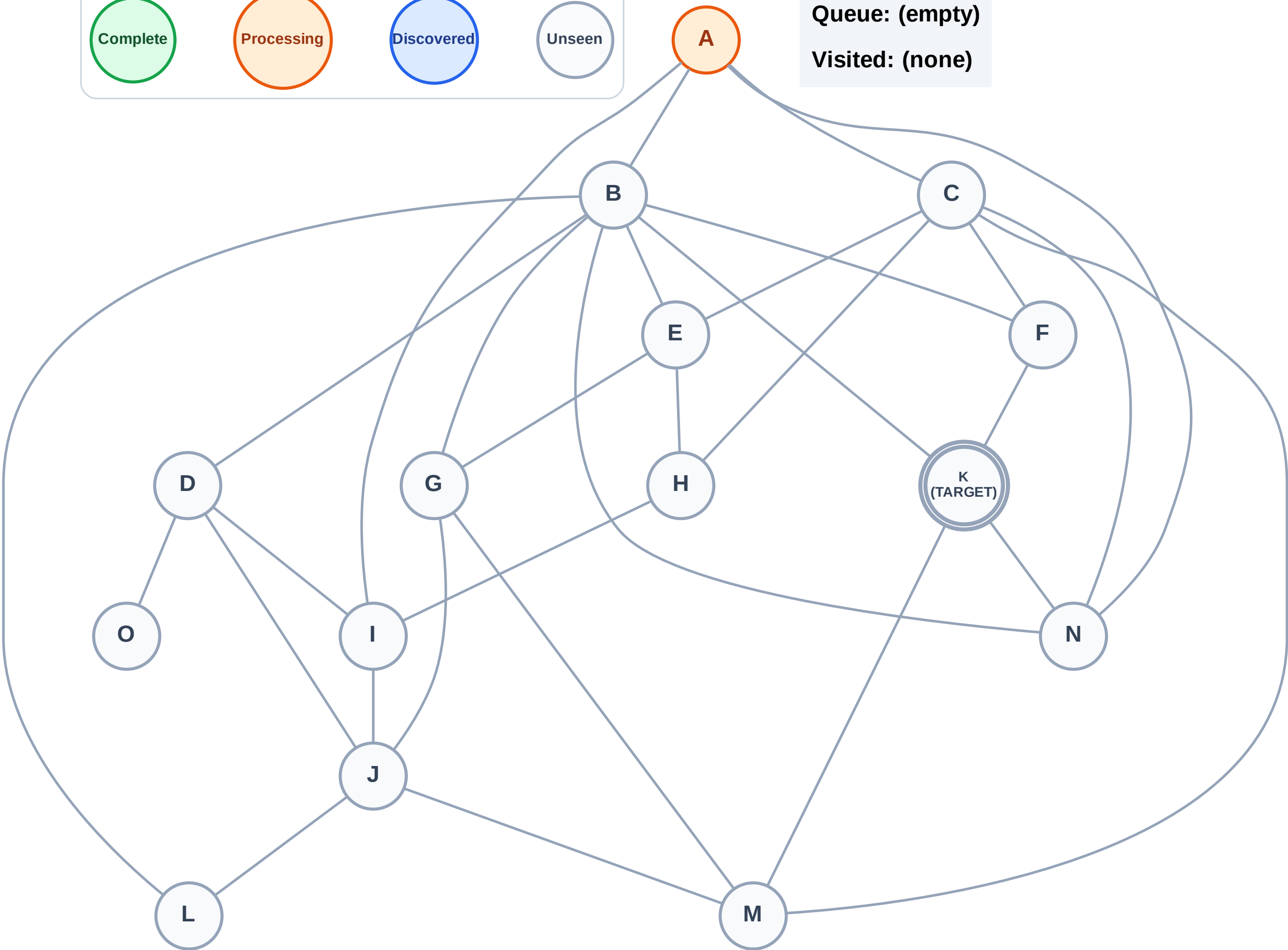
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

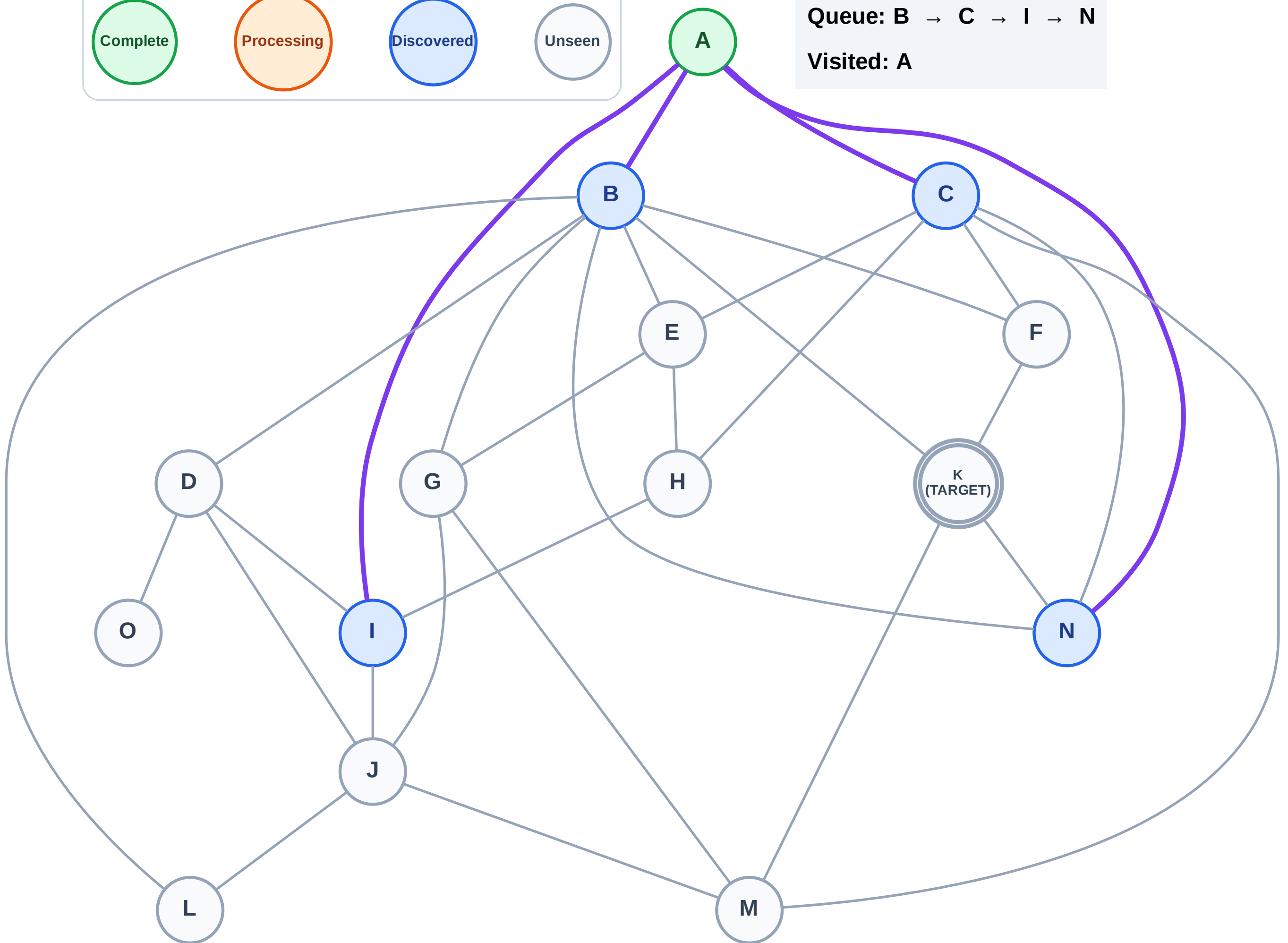
Step 3: Discover B, C, I, N from A

Legend



Queue: B → C → I → N

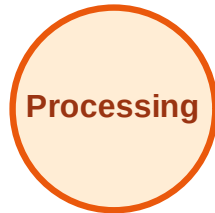
Visited: A



BFS Traversal

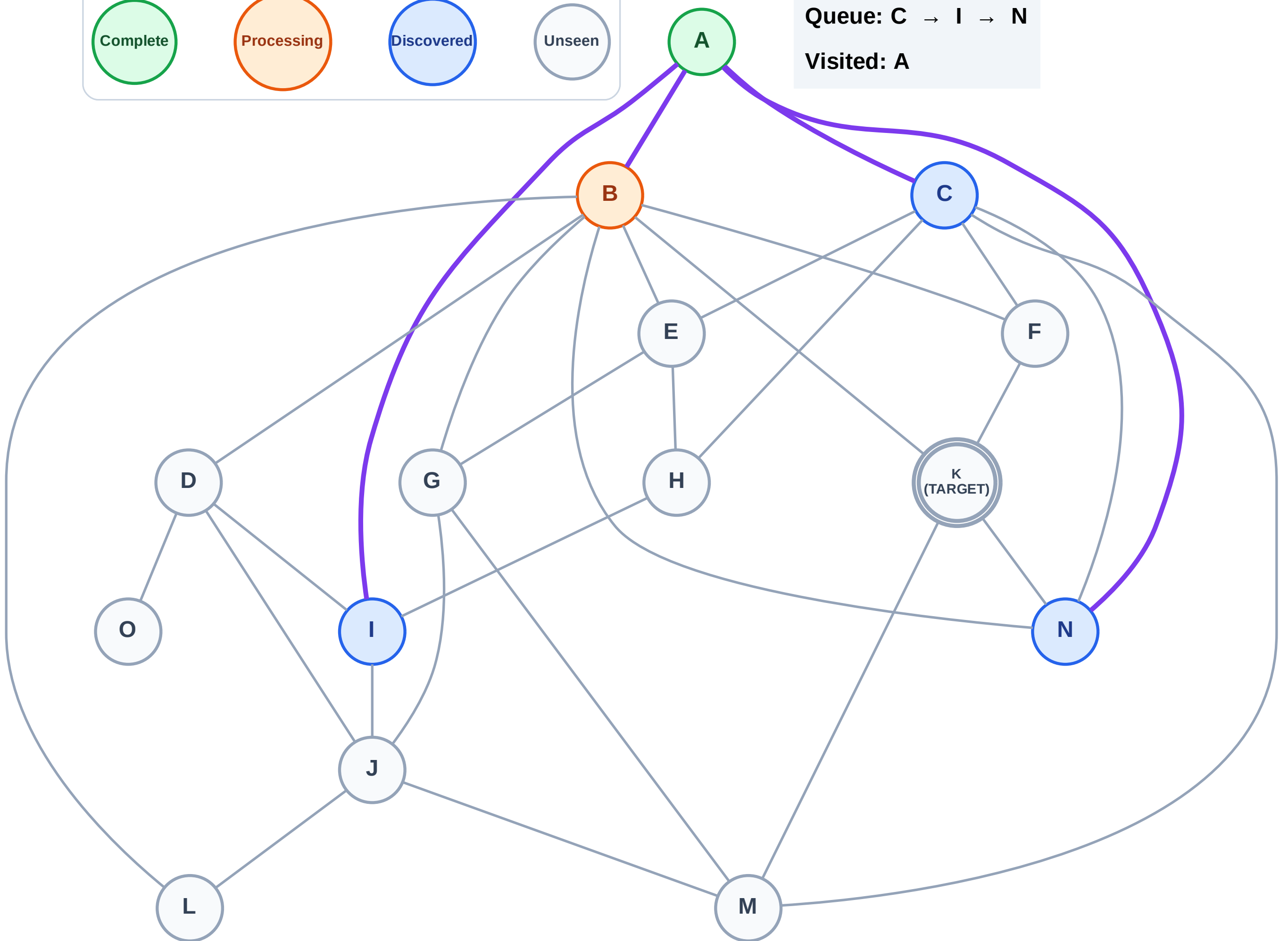
Step 4: Process node B

Legend



Queue: C → I → N

Visited: A



BFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

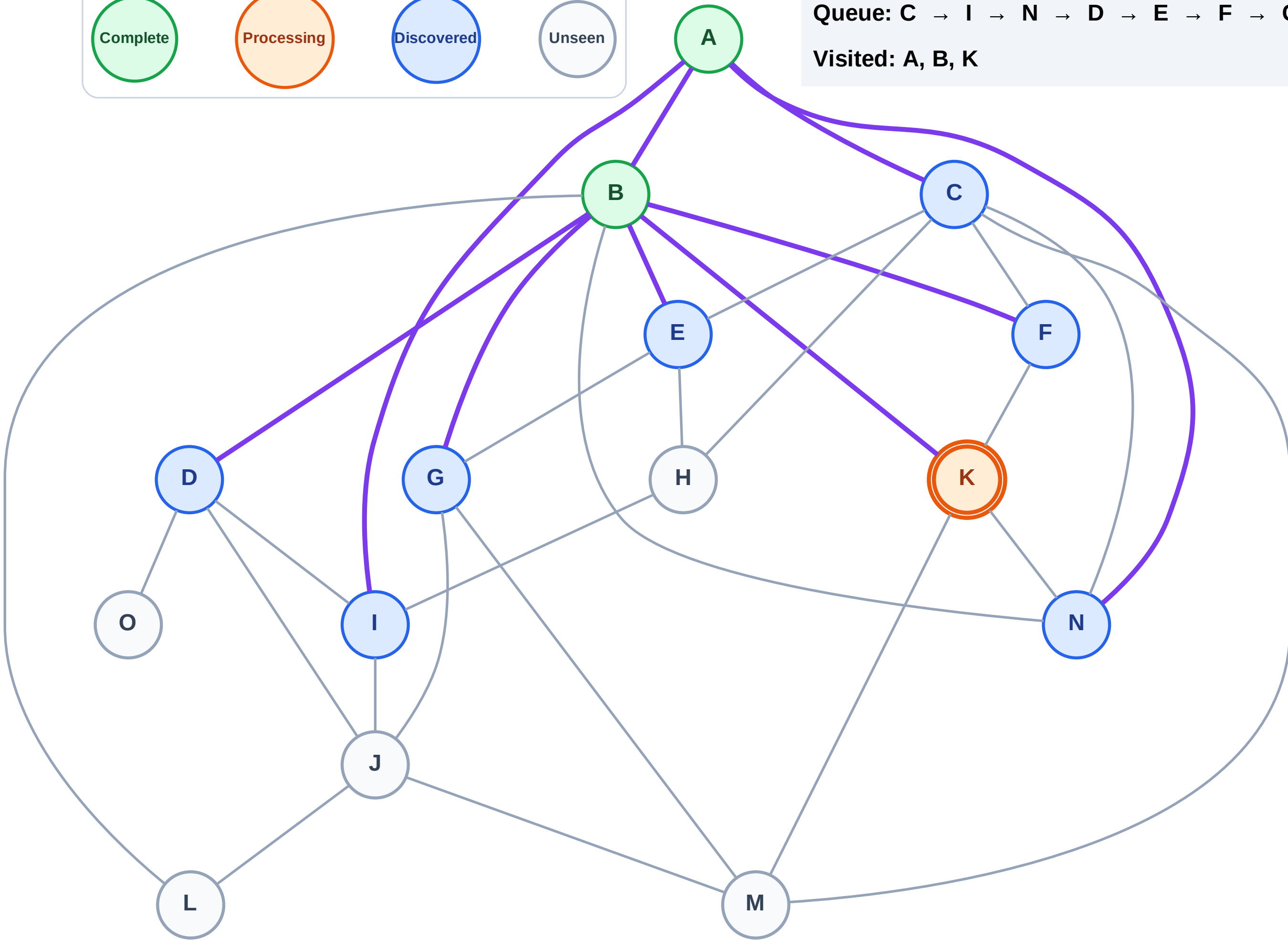
Processing

Discovered

Unseen

Queue: C → I → N → D → E → F → G

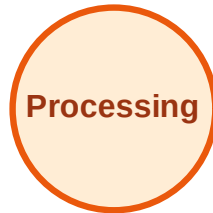
Visited: A, B, K



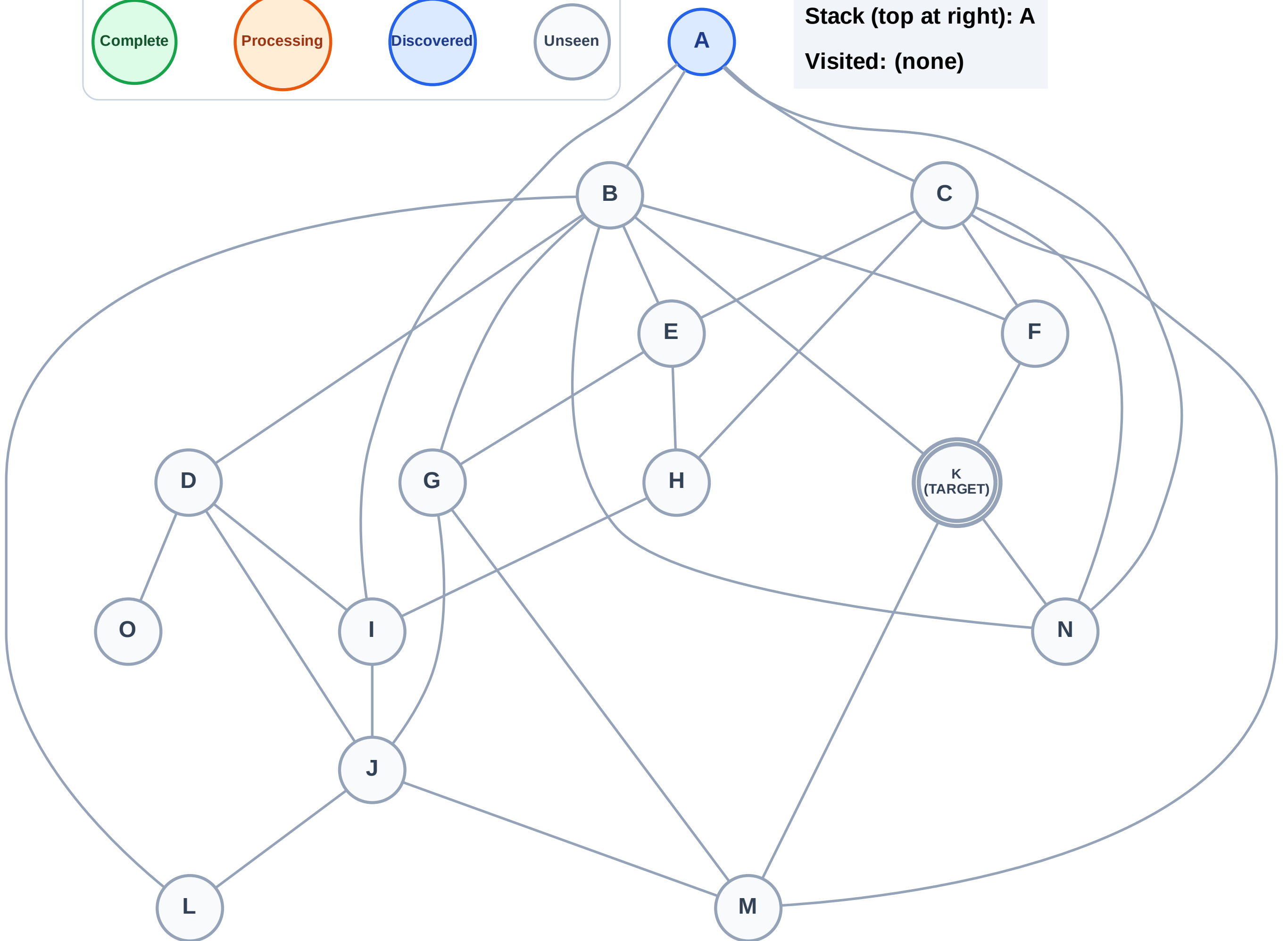
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

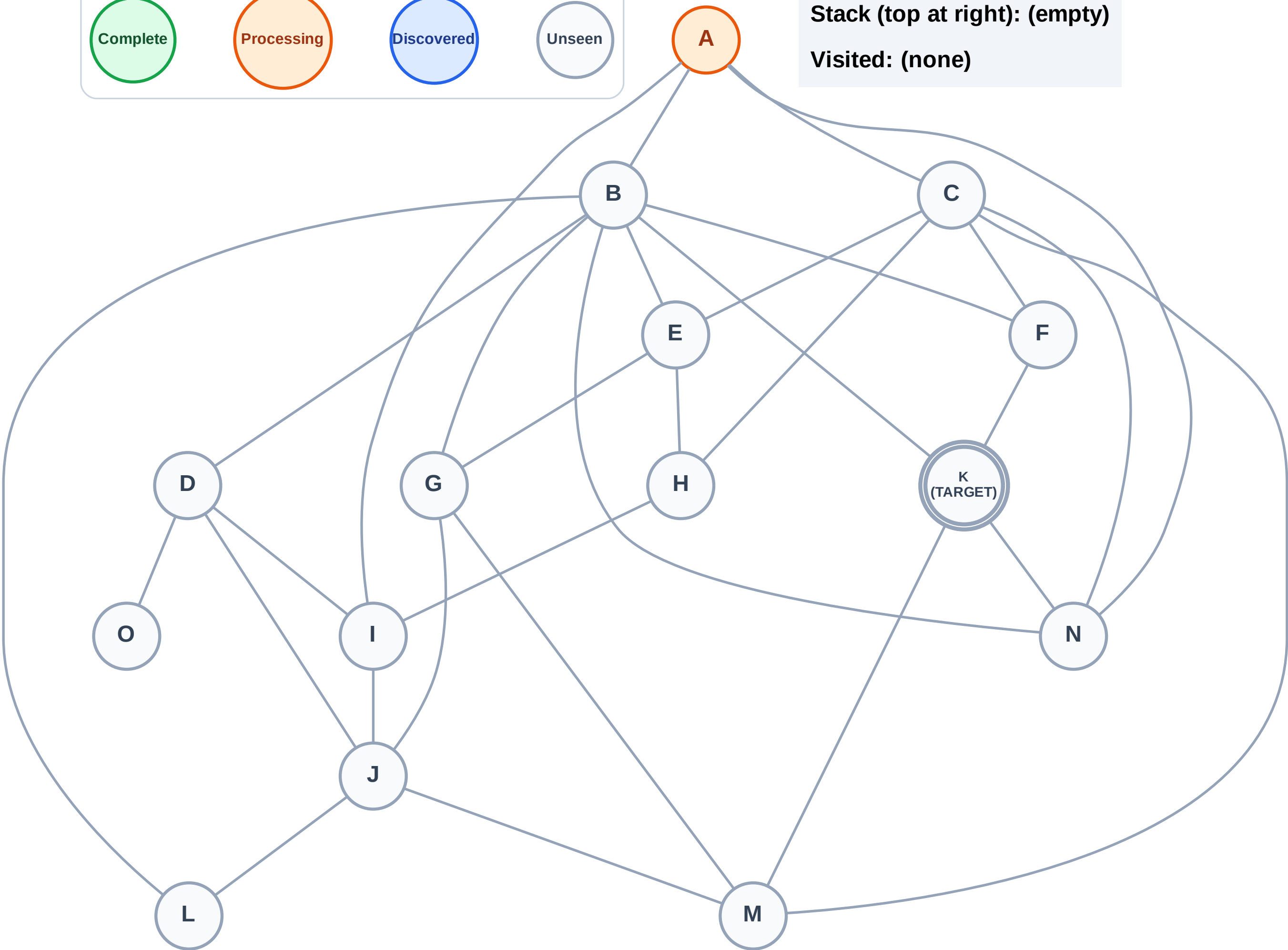
Complete

Processing

Discovered

Unseen

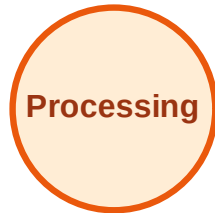
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

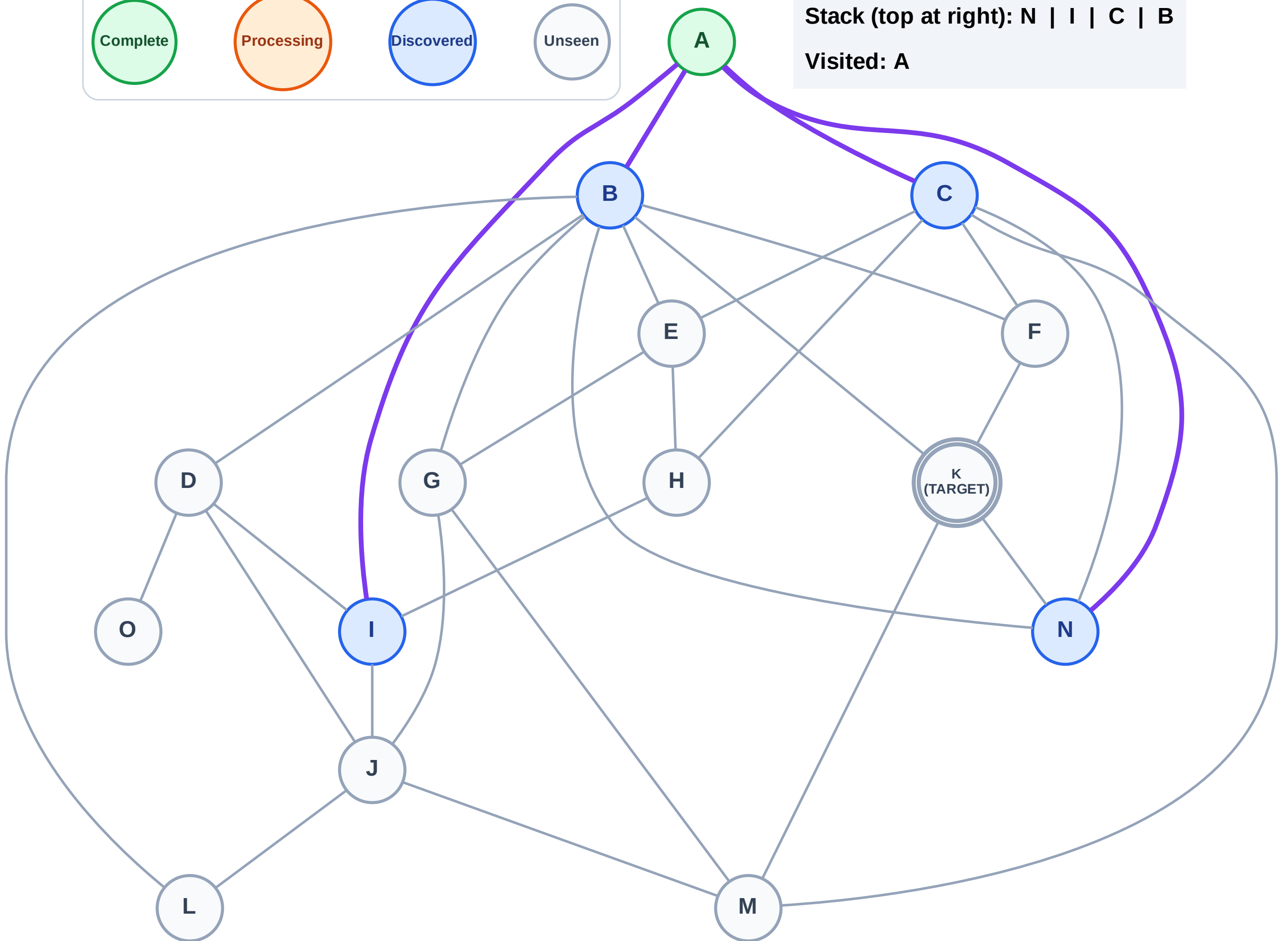
Step 3: Discover N, I, C, B from A

Legend



Stack (top at right): N | I | C | B

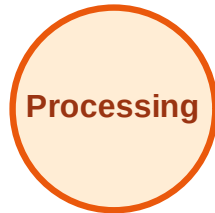
Visited: A



DFS Traversal

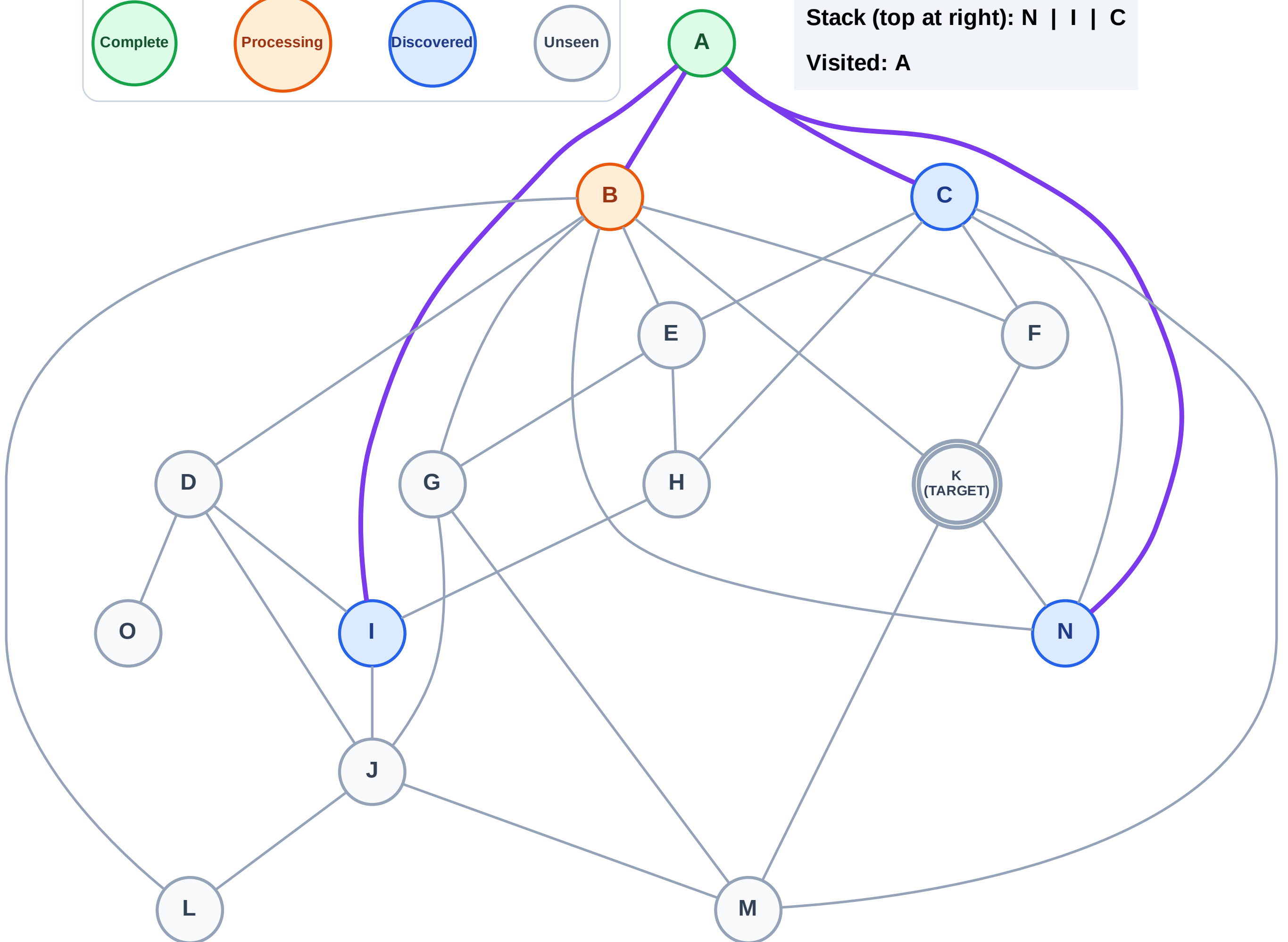
Step 4: Process node B

Legend



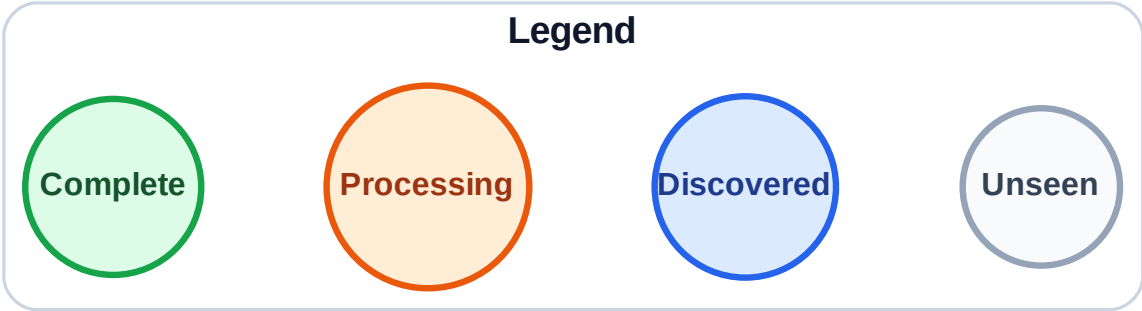
Stack (top at right): N | I | C

Visited: A



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A $\rightarrow$ B $\rightarrow$ K



Stack (top at right): N | I | C | L

Visited: A, B, K

